TRACK · WILDCARD: BEYOND THE OBVIOUS

HemaSight

Needle-free anaemia screening on a smartphone — for women, adolescent girls and young children in low-resource communities.

TEAM
zeroBytes

TEAM LEADER
Ravi Sah

DOMAIN
Healthcare & Public Health

DOCUMENT
Datathon supporting brief

MEMBERS
Ravi Sah · Allaudin Ansari · Bishal Kumar Sah · Vansh Kumar Jain · Aryan Mandlik

01 Problem Statement

Anaemia is one of the world's most widespread yet most neglected health problems — and the main reason it persists is that most people are never tested.

40%
of children aged 6–59 months are anaemic

37%
of pregnant women aged 15–49 are anaemic

30%
of women aged 15–49 are anaemic

That is roughly half a billion women and 269 million young children worldwide (WHO). Progress has been so slow that the world is not on track to halve anaemia in women of reproductive age. In pregnancy, anaemia raises the risk of maternal death, premature birth and low birth weight; in children, it impairs growth, immunity and cognitive development.

The core barrier is detection, not treatment

Iron supplements are cheap and widely available. Confirming who needs them is the hard part.

Testing needs blood
A finger-prick or venous sample, trained staff, and lancets, cuvettes or strips that are often out of stock at community level.

People avoid the test
Fear of needles and the cost of travel keep many women, girls and children from ever being screened.

Cases are found too late
Anaemia is usually discovered only once it is already moderate or severe.

Follow-up breaks down
Paper registers lose track of detected cases, and planners lack timely local data to target programmes.

? **Our question:** can the smartphone a community health worker already carries screen for anaemia at the doorstep — with no needle, no lab and no consumables — and make sure every flagged case reaches treatment?

02 Proposed Solution

HemaSight estimates haemoglobin (Hb) from photos of the inner eyelid, fingernail beds and palm, then manages each flagged case through to treatment. A working prototype already runs on **IBM LinuxONE**.

STEP 1
Capture
Guided photos of eyelid, nails and palm beside a printed colour card.

STEP 2
Analyse
On-device AI estimates Hb (g/dL) and WHO grade in under 5 seconds.

STEP 3
Act
Voice advice in local language; referral if severe or uncertain.

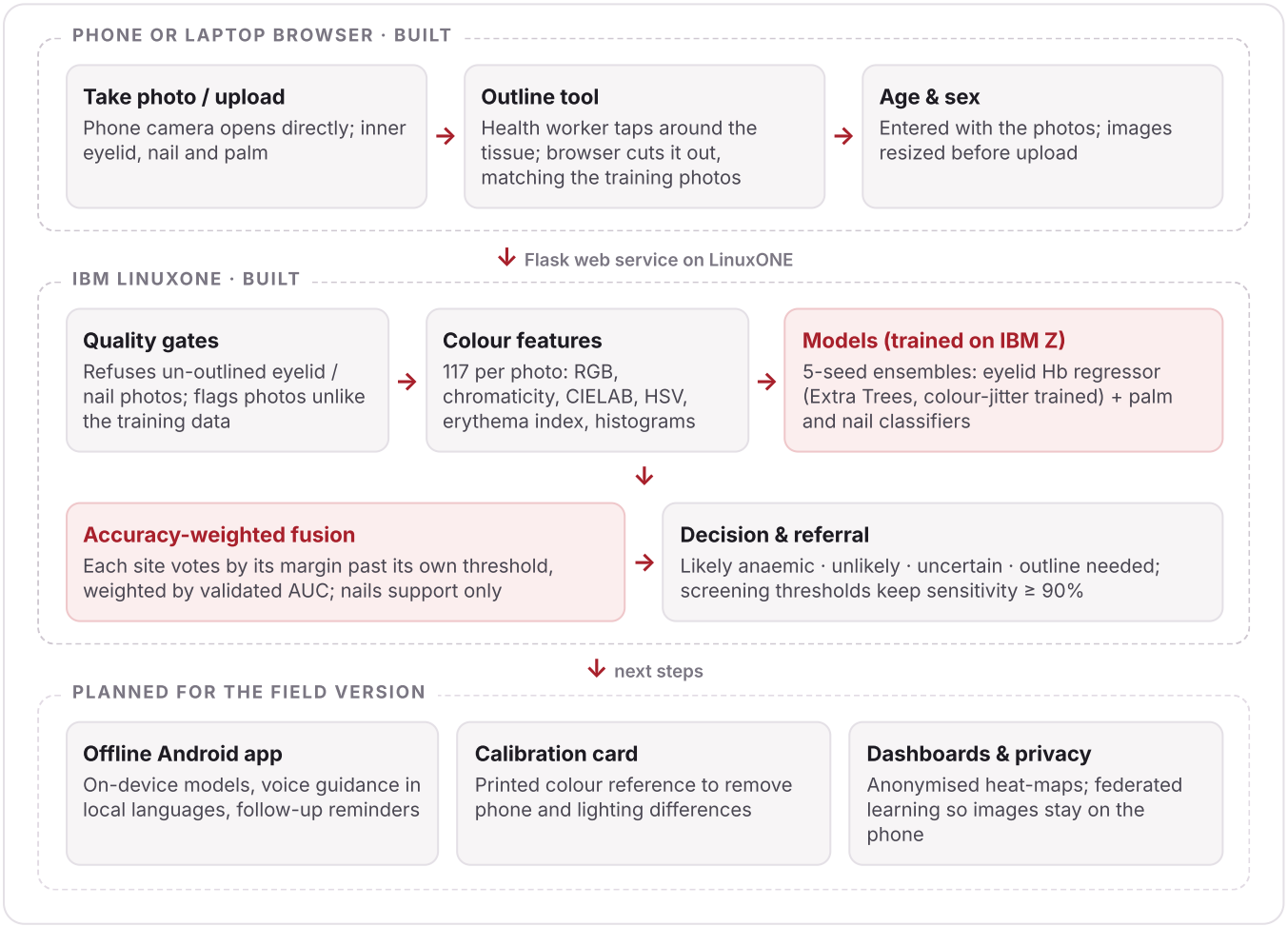
STEP 4
Follow up
Supplement reminders and re-screening at 4–8 weeks.

STEP 5
Plan
Anonymised heat-maps for health and nutrition officials.

A screening tool, not a diagnosis. Every severe or low-confidence result automatically triggers a referral for a confirmatory blood test, so laboratory capacity is spent on the people who need it.

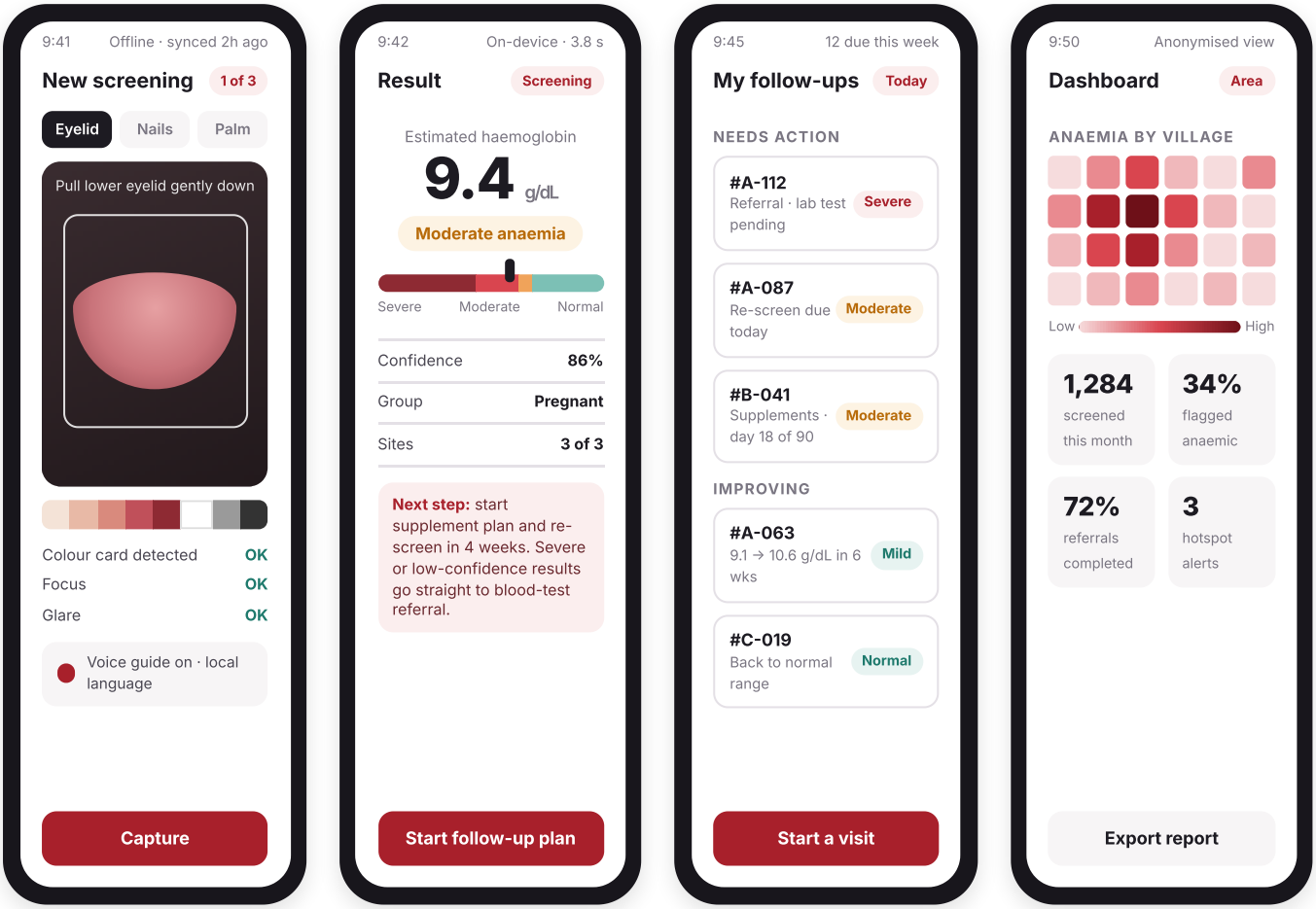
System architecture: built prototype

Everything below is implemented, trained and served on the IBM LinuxONE Community Cloud (s390x).



03 Product Vision: Field App

The full mobile app adds guided capture, follow-up and planning on top of the working screening engine. Four screens cover the journey from the doorstep to the district planning table.



1 · Guided capture
Live checks and a voice guide make photos consistent across phones and lighting.

2 · Instant result
Hb estimate, WHO grade and a confidence score, shown on a simple colour scale.

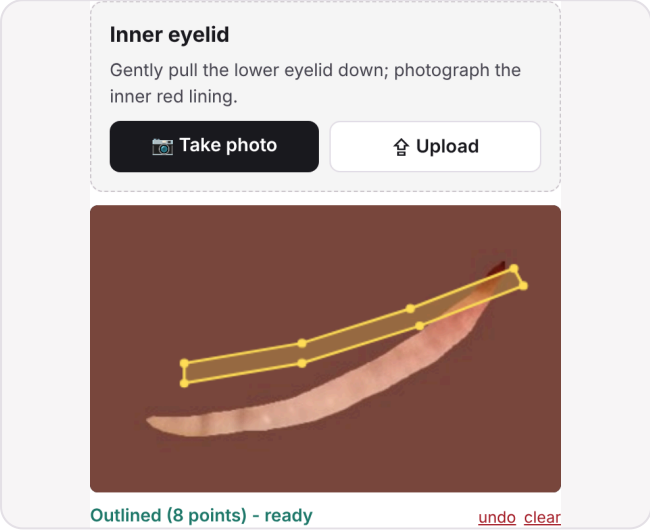
3 · Case follow-up
Every flagged person has a plan, reminders and a re-screen date, so no case is lost.

4 · Planning dashboard
Aggregated heat-maps show where to send supplies and nutrition programmes.

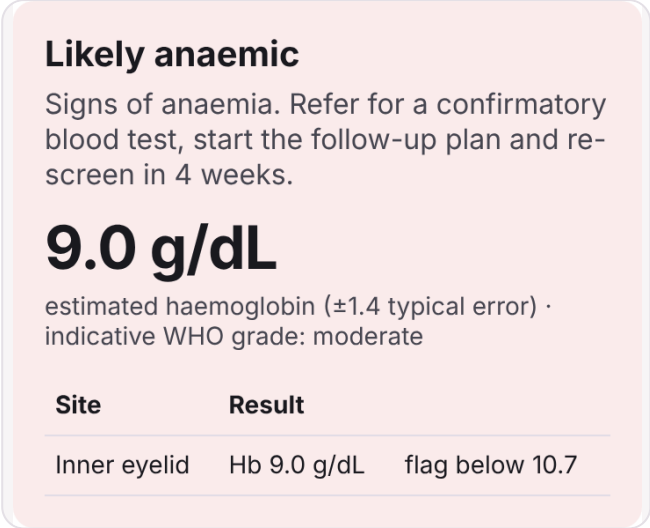
Screens are design concepts; values shown are illustrative, not results. Beneficiaries are identified by code on shared views; names stay on the health worker's device.

04 Working Prototype on IBM LinuxONE

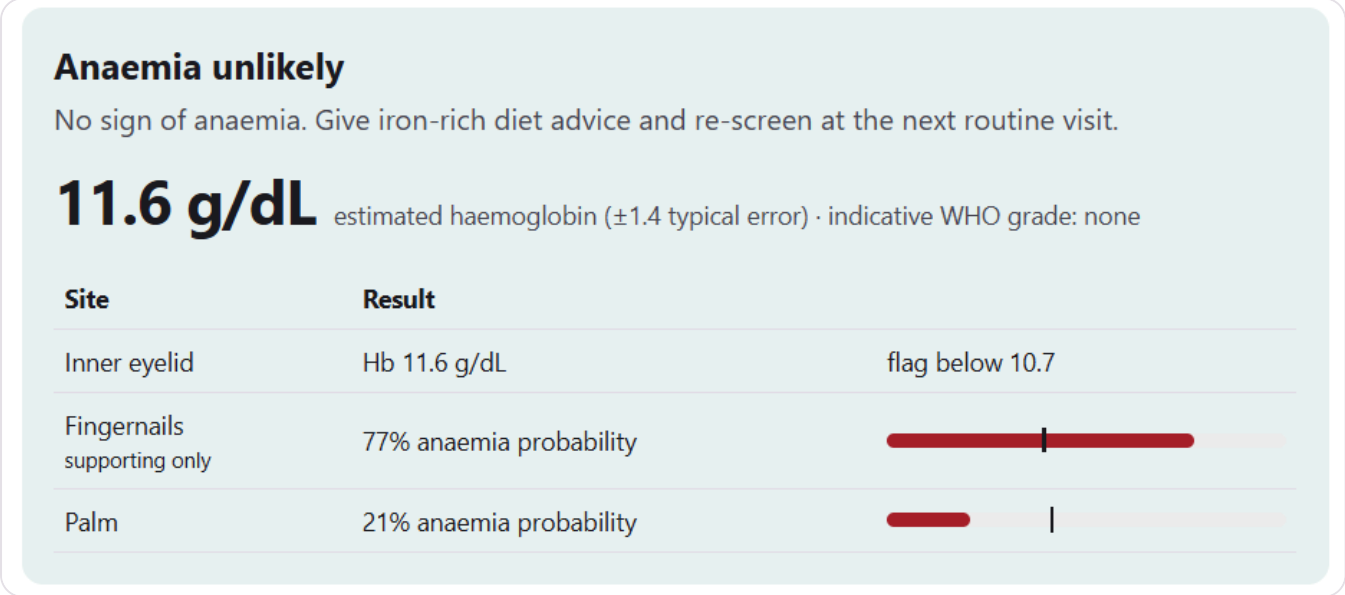
The screening engine is live: a health worker photographs the inner eyelid, palm or nails on any phone, outlines the tissue with a few taps, and gets a result from models trained and served on IBM LinuxONE.



1 · Capture and outline. "Take photo" opens the phone camera. Tapping around the red lining cuts out the tissue exactly like the training photos.



2 · Result. Estimated Hb with typical error, indicative WHO grade, and a clear action: refer for a confirmatory blood test.



3 · Three sites, one decision. Eyelid and palm agree (not anaemic); the nail model disagrees but counts only as supporting evidence, so it cannot overrule them.

Built on IBM Z

Trained on LinuxONE

All models trained on the IBM LinuxONE Community Cloud (s390x, Ubuntu 24.04) with scikit-learn and IBM Snap ML.

Served from LinuxONE

The Flask web app runs on the same server; photos are deleted right after analysis.

Deep learning without PyTorch

PyTorch has no IBM Z build, so we ported ResNet-18 to pure numpy: about 0.2 s per photo on s390x.

05 Results, Measured on IBM LinuxONE

We test the way a real deployment would be judged: on **hospitals the model never saw** (train on 9, test on the 10th), and on photos re-coloured as if taken by a **different phone camera**.

0.93
AUC for anaemia at unseen hospitals (eyelid Hb model, 710 children, 10 hospitals)

90%
of anaemic children flagged, while clearing 79% of healthy ones (estimate < 10.7 g/dL)

0.86
AUC on a simulated new phone camera (±8% colour), vs 0.67 without robustness training

EYELID HB MODEL · UNSEEN HOSPITALS	NORMAL PHOTOS	NEW CAMERA ±4%	NEW CAMERA ±8%
5-seed ensemble, Extra Trees + colour jitter (deployed)	**AUC 0.930** · MAE 1.39 g/dL	**0.906**	**0.856**
Same, single seed (average of 5 seeds)	AUC 0.928 ± 0.003 · MAE 1.40	–	–
Same features, plain training	AUC 0.939 · MAE 1.34	0.816	0.671
IBM Snap ML Random Forest	AUC 0.919 · MAE 1.42	0.889	0.833
Pretrained ResNet-18 (numpy on s390x)	AUC 0.892 · MAE 1.45	–	–
Baseline: always guess the average Hb	MAE 1.81 g/dL	–	–

YES / NO MODELS · 5-SEED ENSEMBLES, GROUPED 5-FOLD CV	NORMAL	NEW CAMERA ±4%	±8%
Palm (Random Forest)	AUC 0.876	0.850	0.818
Fingernails (Extra Trees, supporting only)	AUC 0.775	0.739	0.722

What we learned

- **Robustness beats peak accuracy.** The plain model collapsed on a new camera (0.94 → 0.67 AUC). Training on colour-shifted copies keeps 0.85; a 5-seed ensemble then lifts every score (0.93 / 0.91 / 0.86).
- **Colour beats deep features.** Pallor is a colour signal; ImageNet networks learn to ignore colour, so ResNet-18 scored lower.
- **No leakage.** All augmented copies of a photo stay in the same fold.

Honest limitations

- Average Hb error is 1.39 g/dL, above our 1.0 target; estimates are pulled toward the average, so severe cases read as moderate (still flagged).
- Trained on children aged 6–59 months in Ghana; adults and pregnant women need local validation.
- Large lighting changes still hurt; the calibration card is the next step.

06 Datasets

All datasets below are publicly available for research. **Used** = in the trained prototype; others are planned. Labels marked Hb value support haemoglobin regression; anaemic / not support classification.

DATASET & SOURCE	CONTENTS	HOW HEMASIGHT USES IT
1. CP-AnemiC · Used Mendeley Data · 10.17632/m53vz6b7fx.1 Hb value	710 hand-cut conjunctiva images of children 6–59 months from 10 hospitals, with lab Hb, age and sex	Trains the eyelid Hb regressor; hospital labels enable unseen-hospital validation
2. Conjunctiva images, Ghana · Used Mendeley Data · 10.17632/nt7r8hv2pz.1 anaemic / not	4,263 conjunctiva images (augmented from the originals)	Eyelid yes/no classifier trained for comparison (AUC 0.89); the app uses the Hb model for the eyelid
3. Palm images · Used Mendeley Data · 10.17632/ccr8cm22vz.1 anaemic / not	4,262 palm images of children ≤5 years	Palm classifier; copies of a photo grouped so they never leak between train and test
4. Fingernail images · Used Mendeley Data · 10.17632/2xx4j3kjg2.1 anaemic / not	4,217 nail-bed images (dataset typo "Non-Anrmic" handled)	Nail classifier, supporting evidence only
5. Eyes-defy-anemia IEEE DataPort Hb value	218 eye images with hand-segmented conjunctiva	Planned: train automatic segmentation to replace manual outlining
6. Fingertip video dataset Sabir et al., IEEE Access 2024 Hb value	1-min smartphone fingertip videos, 150 patients, Hb 4.3–12.4 g/dL	Planned: external benchmark for Hb error
7. Demographic & Health Surveys DHS Program · open govt. portals context	Population-level Hb and demographic records; district fact sheets	Planned: baseline heat-maps for officials
8. HemaSight pilot data Own collection, with consent Hb value	Paired eyelid + nail + palm photos with calibration card and lab Hb, from a partner clinic after ethics approval	Fine-tuning and validation on adults, adolescent girls and pregnant women

Data handling rules

- **Grouped splits.** Augmented copies (e.g. "Anemic-001FV (3)") are grouped with their original, so near-duplicates never sit on both sides of a split.
- **Per-site models, fused at decision time.** The sites come from different children, so each has its own model and the votes are combined by validated accuracy.
- **Mind the population gap.** Public data is young children only; dataset 8 adds adults and pregnant women before any deployment.

Validation & fairness

- **Unseen-hospital testing:** every reported Hb number comes from hospitals left out of training.
- **Simulated new cameras:** each model is also scored on photos re-coloured by ±4% and ±8%, and chosen by robustness.
- **Safety first:** thresholds are set for ≥ 90% sensitivity; uncertain or severe results always go to a blood test.

1.39
g/dL MAE achieved (target ≤ 1.0, not yet met)

90%
sensitivity achieved at 79% specificity (target met)

13,496
photos used to train the models on LinuxONE

0.86
AUC on a new phone camera (±8% colour)

07 Who It Serves

1. **Pregnant & lactating women**
   Regular, painless Hb checks to reduce the risk of maternal death, premature birth and low birth weight.
2. **Children, 6 months – 5 years**
   The highest-risk group; parents accept a photo far more readily than a finger-prick.
3. **Adolescent girls, 10–19 years**
   Screening in schools and communities before anaemia affects growth and learning.
4. **Women of reproductive age**
   Quick screening during routine home visits, with no travel or cost.
5. **Community health workers**
   Primary users: a voice-guided tool with no consumables that replaces paper registers.
6. **Primary-care doctors**
   Prioritised referrals, so blood tests go to those who need them.
7. **Health & nutrition officials**
   Anonymised local data and heat-maps to target programmes where anaemia is highest.

Why it is beyond the obvious

Three body sites, one decision
Eyelid, palm and nails vote, weighted by how accurate each proved to be; weak sites can support but never decide.

Screening plus follow-up
It does not stop at a number — it manages the whole path from detection to re-screening.

Built for the last mile
Offline, entry-level phones, voice-guided, and a calibration card that costs a few cents to print.

Tested like the real world
Unseen hospitals and simulated new phone cameras, not just a random split that flatters the score.

Honest about uncertainty
Unusual or un-outlined photos are refused; uncertain or severe results always go to a blood test.

Zero waste
No lancets, strips or cuvettes — no sharps or plastic from millions of tests.

References

1. World Health Organization. *Anaemia* fact sheet. who.int/news-room/fact-sheets/detail/anaemia

2. Stevens GA et al. National, regional and global estimates of anaemia by severity, 2000–19. *Lancet Global Health* 2022;10.

3. Dimauro G et al. *Eyes-defy-anemia* dataset. IEEE DataPort.

4. Appiahene P, Asare JW et al. CP-AnemiC: a conjunctival pallor dataset and benchmark for anaemia detection in children. *Med Novel Technol Devices* 2023;18:100244.

5. Asare JW, Appiahene P, Donkoh E. Palm (10.17632/ccr8cm22vz.1) and fingernail (10.17632/2xx4j3kjg2.1) image datasets. Mendeley Data, 2022.

6. Sabir H, Khan KU, Ishaq O. Fingertip video dataset for non-invasive diagnosis of anaemia. *IEEE Access* 2024.

7. The DHS Program. Demographic and Health Survey datasets. dhsprogram.com

8. Appiahene P, Asare JW, Donkoh E. Conjunctiva image dataset, Ghana. Mendeley Data, 10.17632/nt7r8hv2pz.1, 2022.

9. IBM LinuxONE Community Cloud; IBM Snap ML. ibm.com